

Midterm 1 - Solutions

QUESTION 1

- a. The first experiment is about which beverage we drink. We can denote C to drinking coffee, T to drinking tea, and J to drinking orange juice. We also denote S to falling asleep and W to not falling asleep (or remaining awake). The sample space should contain all possibilities of beverages being drunk and falling or not falling asleep in class. Specifically, we denote the elements of the sample space as a combination of two letters XY , $X \in \{C, T, J\}$ and $Y \in \{S, W\}$. Thus, the sample space is:

$$\Omega = \{CS, CW, TS, TW, JS, JW\}$$

To construct the probability measure, we need to calculate the probability of each element in the sample space. This will give a well-defined probability measure because any event will have a probability given by the sum of the probabilities of each of their elements (axiom 3). Axiom 1 will clearly be satisfied, so the only thing we will need to check is that the probability of Ω is 1.

We know that:

$$\begin{aligned}\mathbb{P}(C) &= 0.5 = \frac{1}{2}, \\ \mathbb{P}(T) &= 0.4 = \frac{4}{10} = \frac{2}{5}, \\ \mathbb{P}(J) &= 0.1 = \frac{1}{10}.\end{aligned}$$

Given the drink, we know the probability of falling asleep is:

$$\begin{aligned}\mathbb{P}(S|C) &= 0.1 = \frac{1}{10}, \\ \mathbb{P}(S|T) &= 0.2 = \frac{2}{10} = \frac{1}{5}, \\ \mathbb{P}(S|J) &= 0.8 = \frac{8}{10} = \frac{4}{5}.\end{aligned}$$

Using the definition of the conditional expectation, we can calculate the probability of each element in the sample space:

$$\begin{aligned}\mathbb{P}(CS) &= \mathbb{P}(C \cap S) = \mathbb{P}(S|C)\mathbb{P}(C) = \frac{1}{10} \cdot \frac{1}{2} = \frac{1}{20} = 0.05, \\ \mathbb{P}(CW) &= \mathbb{P}(C \cap W) = \mathbb{P}(W|C)\mathbb{P}(C) = \mathbb{P}(S^c|C)\mathbb{P}(C) = (1 - \mathbb{P}(S|C))\mathbb{P}(C) = \frac{9}{10} \cdot \frac{1}{2} = \frac{9}{20} = 0.45, \\ \mathbb{P}(TS) &= \mathbb{P}(T \cap S) = \mathbb{P}(S|T)\mathbb{P}(T) = \frac{1}{5} \cdot \frac{2}{5} = \frac{2}{25} = 0.08, \\ \mathbb{P}(TW) &= \mathbb{P}(T \cap W) = \mathbb{P}(W|T)\mathbb{P}(T) = \frac{4}{5} \cdot \frac{2}{5} = \frac{8}{25} = 0.32, \\ \mathbb{P}(JS) &= \mathbb{P}(J \cap S) = \mathbb{P}(S|J)\mathbb{P}(J) = \frac{4}{5} \cdot \frac{1}{10} = \frac{2}{25} = 0.08, \\ \mathbb{P}(JW) &= \mathbb{P}(J \cap W) = \mathbb{P}(W|J)\mathbb{P}(J) = \frac{1}{5} \cdot \frac{1}{10} = \frac{1}{50} = 0.02.\end{aligned}$$

We check that the sum of these gives 1:

$$\mathbb{P}(\Omega) = \frac{1}{20} + \frac{9}{20} + \frac{2}{25} + \frac{8}{25} + \frac{2}{25} + \frac{1}{50} = \frac{10}{20} + \frac{12}{25} + \frac{1}{50} = 1.$$

We showed in class that for a finite sample space, this will define a probability measure.

- b. We need to calculate the probability of drinking juice given that we fall asleep. We use Bayes to get:

$$\mathbb{P}(J|S) = \frac{\mathbb{P}(S|J)\mathbb{P}(J)}{\mathbb{P}(S)}$$

We already know the two probabilities in the numerator so we only need to find $\mathbb{P}(S)$. Notice that $S = \{CS, TS, JS\}$. By axiom 3,

$$\mathbb{P}(S) = \mathbb{P}(CS) + \mathbb{P}(TS) + \mathbb{P}(JS) = \frac{1}{20} + \frac{2}{25} + \frac{2}{25} = \frac{5+8+8}{100} = \frac{21}{100} = 0.21.$$

So we can calculate:

$$\mathbb{P}(J|S) = \frac{\mathbb{P}(S|J)\mathbb{P}(J)}{\mathbb{P}(S)} = \frac{\frac{4}{5} \cdot \frac{1}{10}}{\frac{21}{100}} = \frac{8}{21} \approx 0.38.$$

Grading comments:

- One possible mistake in part a) was to define different sample spaces for the experiments of which drink we make and for the fact that we fall asleep or not, instead of a sample space for the result of the joint experiment. Less common was to add the day of the week on the sample space.
- Another possible mistake was to forget to calculate the probabilities for each element of the sample space or not mention why it is enough to consider the probabilities over the singletons.
- For part b) a possible mistake was to calculate $\mathbb{P}(S|J^c) = \mathbb{P}(S|C) + \mathbb{P}(S|T)$. This doesn't hold. We have that:

$$\mathbb{P}(S|J^c) = \frac{\mathbb{P}(S \cap J^c)}{\mathbb{P}(J^c)} = \frac{\mathbb{P}(CS) + \mathbb{P}(TS)}{1 - \mathbb{P}(J)} = \frac{0.13}{0.9} = \frac{13}{90}$$

On the other hand:

$$\mathbb{P}(S|C) + \mathbb{P}(S|T) = \frac{\mathbb{P}(CS)}{\mathbb{P}(C)} + \frac{\mathbb{P}(TS)}{\mathbb{P}(T)} = \frac{0.05}{0.5} + \frac{0.08}{0.4} = \frac{5}{50} + \frac{8}{40} = \frac{3}{10}.$$

QUESTION 2

- a. Let X be the number of dollar the Chevalier wins in one game. Then we know that

$$\mathbb{P}(X = 1) = \mathbb{P}(X = -2) = \mathbb{P}(X = 3) = \mathbb{P}(X = -4) = \mathbb{P}(X = 5) = \mathbb{P}(X = -6) = \frac{1}{6}.$$

As a result, the expectation can be calculated as

$$\mathbb{E}[X] = \frac{1 - 2 + 3 - 4 + 5 - 6}{6} = -0.5.$$

- b. For a single round, we know that $\mathbb{P}(X < 0) = 0.5$. Therefore, the number of rounds N follows binomial distribution $\text{Binom}(5, 0.5)$. So we have

$$\mathbb{P}(N = 3) = \binom{5}{3} (0.5)^3 (0.5)^2 = \frac{5}{16}.$$

Grading comments:

- A common mistake in part (a) is misunderstanding the concept of expectation. Some students failed to apply the formula in Definition 1.8.1 correctly. Some of them failed to determine the value of $\varphi(i)$. In their solutions, $\varphi(i)$ is not deterministic, and some random variables are still involved in it. Some students try to use the law of total expectation but didn't get it right.
- In part (b), there is no common mistake. One point is subtracted if $\binom{5}{3}$ is not explicitly calculated.

QUESTION 3

- a. Let N be the day the avocado ripens. Since each day, the avocado ripens independently of other days, and each day, it has the same probability of $\frac{1}{10}$ to ripen, we conclude that N is a geometric random variable with probability of success $p = \frac{1}{10}$. Therefore,

$$\mathbb{E}[N] = 1/p = 10.$$

- b. Label the avocados 1, ..., 5 and let N_i be the day the i -th avocado ripens. Let X_i be the indicator random variable that the i -th avocado is ripe on the day of the super bowl. Thus,

$$X = \sum_{i=1}^5 X_i$$

is the number of avocados that are ripe on the day of the super bowl. Now, as clarified in the problem, the i -th avocado is ripe on the day of the super bowl if and only if the i -th avocado ripens on the third possible day. Also, as determined in a), N_i is a geometric random variable. Therefore,

$$\mathbb{P}(X_i = 1) = \mathbb{P}(N_i = 3) = \left(1 - \frac{1}{10}\right)^2 \frac{1}{10} = 0.081$$

Since the avocados ripen independently, the X_i 's are independent and have the same probability of success. Therefore, X is a binomial(5, .081) random variable, so

$$\mathbb{E}[X] = 5(0.081) = 0.405.$$

Alternative Solution: Let Z_i denote the number of avocados that are ripe on day i for $i=1,2,3$. Now, the avocados ripen independently on the first day with probability .1, so Z_1 is a binomial(5,.1) random variable. After the first day, there are $5 - Z_1$ avocados remaining, and the remaining avocados ripen independently with probability .1, so the conditional distribution of Z_2 given Z_1 is binomial($5 - Z_1, 0.1$). Likewise, the conditional distribution of Z_3 given Z_1 and Z_2 is binomial($5 - Z_1 - Z_2, 0.1$). We seek to calculate

the expectation of Z_3 , so using the law of total probability:

$$\begin{aligned}
\mathbb{E}[Z_3] &= \sum_{x=1}^5 \sum_{y=1}^x \mathbb{E}[Z_3|Z_2 = y, Z_1 = x] \mathbb{P}(Z_2 = y, Z_1 = x) \\
&= \sum_{x=1}^5 \sum_{y=1}^x (0.1)(5 - x - y) \mathbb{P}(Z_2 = y, Z_1 = x) \\
&= \sum_{x=1}^5 \sum_{y=1}^x (0.1)(5 - x - y) \mathbb{P}(Z_2 = y, Z_1 = x) \\
&= 5(0.1) \sum_{x=1}^5 \sum_{y=1}^x \mathbb{P}(Z_2 = y, Z_1 = x) - (0.1) \sum_{x=1}^5 \sum_{y=1}^x y \mathbb{P}(Z_2 = y, Z_1 = x) \\
&\quad - (0.1) \sum_{x=1}^5 \sum_{y=1}^x x \mathbb{P}(Z_2 = y, Z_1 = x)
\end{aligned}$$

Now, the first sum is just summing over all possible values of Z_1 and Z_2 , so we get

$$\sum_{x=1}^5 \sum_{y=1}^x \mathbb{P}(Z_2 = y, Z_1 = x) = 1.$$

We can rewrite the second sum using conditional probabilities

$$\begin{aligned}
\sum_{x=1}^5 \sum_{y=1}^x y \mathbb{P}(Z_2 = y, Z_1 = x) &= \sum_{x=1}^5 \left(\sum_{y=1}^x y \mathbb{P}(Z_2 = y|Z_1 = x) \right) \mathbb{P}(Z_1 = x) \\
&= \sum_{x=1}^5 \mathbb{E}[Z_2|Z_1 = x] \mathbb{P}(Z_1 = x) \\
&= \sum_{x=1}^5 (5 - x)(0.1) \mathbb{P}(Z_1 = x) \\
&= 5 - \mathbb{E}[Z_1] = (0.1)(5 - (5)(0.1)) = 0.45
\end{aligned}$$

Likewise, the third sum can be simplified

$$\begin{aligned}
\sum_{x=1}^5 \sum_{y=1}^x x \mathbb{P}(Z_2 = y, Z_1 = x) &= \sum_{x=1}^5 x \sum_{y=1}^x \mathbb{P}(Z_2 = y, Z_1 = x) \\
&= \sum_{x=1}^5 x \mathbb{P}(Z_1 = x) = \mathbb{E}[Z_1] = (5)(0.1) = 0.5.
\end{aligned}$$

So, as a final answer, we obtain:

$$\mathbb{E}[Z_3] = (5)(0.1)(1) - (0.1)(0.45) - (0.1)(0.5) = 0.405$$

- c. Let Y be the number of friends that show up, and X be as in part b). The problem tells us that X is independent of Y and that Y is uniform on $\{0, \dots, 5\}$. The event that we have exactly the same number of friends showing up as ripe avocados is the event that $X = Y$. Then, by the law of total probability,

$$\mathbb{P}(X = Y) = \sum_{n=0}^5 \mathbb{P}(X = Y|Y = n) \mathbb{P}(Y = n) = \frac{1}{6} \sum_{n=0}^5 \mathbb{P}(X = n|Y = n)$$

$$= \frac{1}{6} \sum_{n=0}^5 \mathbb{P}(X = n) = \frac{1}{6}$$

where in the second to last equality we used independence, and in the last equality we used the fact that X only takes values in $\{0, \dots, 5\}$.

Grading comments:

- The most common mistake was in part b) where some people calculated $\mathbb{P}(N_i \leq 3)$ instead of $\mathbb{P}(N_i = 3)$
- There were multiple people who attempted to do part b) by the second given method but said Z_1 was $\text{binomial}(5 - \mathbb{E}[Z_1], 0.1)$. A binomial distribution must have a whole number as its first parameter, so this is not allowed, and we must use conditional distributions.
- In part c), it was common for people to say that Y was uniform on $\{1, \dots, 5\}$ instead of $\{0, \dots, 5\}$